Multimedia Signal Processing - 2nd Module

March, 3 2017

1. **Question (8 pts)**
   Consider the following sequence of samples:

$$x = \{14,\ 15,\ 18,\ 16,\ 22,\ 17,\ 18,\ 13,\ 13\}.$$

   (a) Obtain the sequence $\tilde{x}$ by encoding and decoding x with a closed loop DPCM. Highlight which is the encoded signal, and which is the decoded version. Assume the first sample is encoded without loss, and residuals are quantized according to the following rules:

$$\begin{aligned} I_1 &= [-\infty, -1.5), & y_1 &= -3 \\ I_2 &= [-1.5, 1.5), & y_2 &= 0 \\ I_3 &= [1.5, +\infty), & y_3 &= 3 \end{aligned}$$

   where I represent decision intervals and y are reproduction levels.

   (b) Compute the mean squared error (MSE) introduced by the use of DPCM.

   (c) Estimate the pmf of each symbol of $\tilde{x}$ and derive the Huffman coding scheme.

   (d) Compute the average code length of: (i) the Huffman code for $\tilde{x}$; (ii) the Shannon code for $\tilde{x}$. Does Huffman code reach the theoretical lower bound?

   (e) Obtain the sequence $\hat{x}$ using an open-loop DPCM on x. Use same parameters of step (a).

   (f) Compute the mean squared error (MSE) introduced by the use of the open loop DPCM.

   (g) Compare results obtained in point (b) and (f), commenting what you see.

2. **Question (5 pts)**
   (a) Explain which are the common visual irrelevancy and redundancy effects that can be exploited for image coding. Discuss them providing some examples.

   (b) Illustrate the block diagram of the JPEG compression coding scheme, commenting each block in details.

3. **Question (5 pts)**
   (a) Introduce the concept of scalar quantization, providing the definition of scalar quantizer and showing the difference between a midtread and a midrise one.

   (b) Describe the compander/expander operation, highlighting how quantization can benefit from it.

4. **Questions (5 pts - each answer can be either TRUE or FALSE)**

   (a) Consider a hybrid DCT/DPCM video coding scheme.

   - [T] [F] Only decoders perform motion compensation.
   - [T] [F] Each coding standard strictly defines all operations performed by the encoder.
   - [T] [F] Motion vectors can be computed with sub-pixel accuracy.
   - [T] [F] Motion vectors for one frame can be computed from either past or future frames.

   (b) Consider a vocoder exploiting source-system model to encode speech signal.

   - [T] [F] Direct quantization of LPC coefficients can lead to instability.
   - [T] [F] LPC coefficients are only estimated for voiced frames
   - [T] [F] Train of pulses can be used as excitation signals for voiced frames, and their period is estimated by the encoder.

   (c) Consider two discrete memoryless sources X and Y emitting sequences of samples from a finite alphabet.

   - [T] [F] Mutual information $I(X;Y)$ is zero for uncorrelated sources.
   - [T] [F] Entropy $H(X)$ is always greater than $H(Y)$ if the alphabet of X is greater than the alphabet of Y.
   - [T] [F] It is possible to lossy encode symbols of X using less than $H(X)$ bits per symbol.

5. **Question (Matlab - 10pts)**

   Consider the audio file `gb.wav`.

   (a) Load 1 second of the file into the variable `y`. Draw the loaded waveform in a figure expressing the x-axis in seconds.

   (b) Encode `y` with a scalar uniform quantizer (PCM) whose levels are represented with $R = 8$ bits. Adapt levels to the audio dynamic.

   (c) Repeat step (b) using $R = [4, 8, 12, 16, 20]$ and draw the rate distortion curve. Use SNR in dB as distortion metric.

   (d) Encode `y` using KLT, considering blocks of $B = 4$ samples. Use a scalar uniform quantizer whose levels are represented by $R = 8$ bits. Adapt levels to the dynamic of the signal to quantize. Use the same quantizer for all coefficients.

   (e) Decode the signal obtained in the previous point using inverse KLT.

   (f) Compute the distortion introduced by KLT by means of SNR in dB.

   (g) Compare KLT and PCM distortion results. Explain how to improve KLT results with respect to the ones obtained.